

FIT5225 Assignment 2

Individual Report

Junbo Chen | Student ID 36970271

Name: Junbo Chen

Student ID: 36970271

Repository: <https://github.com/chenjunboa/PacificBioArchive>

Role and Contribution

My role was to turn the assignment brief into the first complete local prototype and establish a stable foundation that the other three members could extend in sequence. I designed the initial multi-cloud architecture, repository layout, API contracts and ownership boundaries. I implemented the baseline FastAPI service, local JWT authentication, upload checksum validation and duplicate handling, local media storage, query and tag flows, deletion behaviour, and the initial React user interface. I also created the inference service scaffold, Terraform structure, Docker Compose local environment, automated tests, and the Chinese handoff documents used by later members.

My main objective was to make the project runnable before cloud deployment rather than leave each member with disconnected pieces. I verified the local API and interface, added representative image and video processing paths, and documented exact responsibilities for AWS/GCP deployment, backend completion and final UI integration. At the end of the project I returned as the release reviewer, checked the latest cloud evidence and CI results, fixed a Windows-specific Playwright startup issue, added the official-icon architecture diagram, and approved the tested commit for the release-candidate-1 tag.

Reflection

The project demonstrated that sequential ownership can work for a cloud assignment only when each handoff contains an executable baseline, clear acceptance criteria and evidence. Building the first prototype gave the team a shared vocabulary for media states, checksums, tags, ownership and query behaviour. It also reduced uncertainty for the deployment stage because later members could compare cloud behaviour with a known local workflow.

The main weakness of the sequential approach was that integration risks appeared late. Some cloud paths, indexes and worker routing decisions could not be proven until the final interface exercised them together. I learned to separate “implemented in the repository” from “verified in the deployed system”, and to record unresolved items instead of assuming that a handoff had completed them. The final review reinforced the value of cross-platform E2E tests: the workflow passed in Linux CI but its local startup command initially failed on Windows. In a future project I would define the

cloud evidence checklist and final-report ownership at the beginning, and schedule a shared integration review before the final day.

Generative AI Use

I used OpenAI Codex to help analyse the assignment requirements, structure the prototype, generate focused implementation suggestions, identify edge cases, run and interpret tests, prepare handoff documentation, and review the release candidate. I inspected and tested the resulting work rather than accepting generated output directly. Verification included Python tests, Ruff, Terraform validation, the React production build, Playwright E2E, GitHub Actions and the team's redacted live AWS/GCP smoke evidence. Credentials, verification codes, tokens, presigned URL parameters, Terraform state and model weights were not committed.